This is an area of mathematics known as **kinematics**, which is about the motion of objects.

The derivative gives the slope of a function. It also gives the rate of change of one variable with respect to another variable. In this section we will study **average and instantaneous rates of change** and **motion in a line**.

I- Velocity

Investigation

A diver jumps from a platform at time $t = 0$ seconds. The distance of the diver above water level at time t is given by :

$$s(t) = -4.9t^2 + 4.9t + 10,$$

where s is in metres.

1. Find the **average velocity** of the diver over the given time intervals :

   i) [1,2] ii) [1.5,1] iii) [1.1,1] iv) [1.01,1].

2. Find the **instantaneous velocity** of the diver at $t = 1$ second.

Remark. The average rate of change of s, or average velocity, is the slope of a secant line :

$$\frac{s(t+h) - s(t)}{(t+h) - t} = \frac{s(t+h) - s(t)}{h}.$$

The instantaneous rate of change of s, or the velocity, is the slope of a tangent line :

$$v(t) = \lim_{h \to 0} \frac{s(t+h) - s(t)}{h} = s'(t).$$

Example

During one month, the temperature of the water in a pond is modeled by the function

$$C(t) = 20 + 9te^{-\frac{t}{3}},$$

where t is measured in days and C is measured in degrees Celsius.

1. Find the **average rate of change** in temperature in the first 15 days of the month.
2. Find the **rate of change** in temperature on day 15.

Definition

If an object moves along a straight line, its position from an origin at any time t can be modeled by a **displacement function**, $s(t)$.

Definition

The instantaneous rate of change of displacement is the **velocity function** :

$$v(t) = \lim_{h \to 0} \frac{s(t+h) - s(t)}{h} = s'(t).$$

Property

or $v(t) > 0$, the object is moving to the right or up.

For $v(t) < 0$, the object is moving down or to the left.

For $v(t) = 0$, the object is at rest.

Example

A particle moves in a straight line with a displacement of s metres t seconds after leaving a fixed point. The displacement function is given by

$$s(t) = 2t^3 - 21t^2 + 60t + 3, \quad t \geqslant 0.$$

1. Find the velocity of the particle at any time t.
2. Find the initial position and initial velocity of the particle.
3. Find when the particle is at rest.
4. Find when the particle is moving left and when the particle is moving right.
5. Draw a motion diagram for the particle.

II- Acceleration

Definition

The instantaneous rate of change of the velocity is the **acceleration function** :

$$a(t) = \lim_{h \to 0} \frac{v(t+h) - v(t)}{h} = v'(t) = s''(t).$$

Property

or $a(t) > 0$, the velocity of the object is increasing. For $a(t) < 0$, the velocity of the object is decreasing. For $a(t) = 0$, the velocity is constant.

Remark. Speed is the absolute value of velocity. Velocity tells us how fast an object is moving and the direction in which it is moving.

Speed tells us only how fast it is moving. To determine if an object in motion is speeding up or slowing down you can compare the signs of velocity and acceleration.